

MAT 523: Functions of a Real Variable I

Homework VIII

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(1.)

Problem

We showed in class that if f is measurable, then the set $\{f = \alpha\}$ is measurable for all $\alpha \in \mathbb{R}$. Give an example to show that the converse is false.

Solution

(2.)

Statement

There is a strictly increasing continuous function which takes a set of positive measure to a set of zero measure.

Proof

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(3.)

Statement

Let S be any dense subset of $\mathbb{R}$. Let $f : E \rightarrow \mathbb{R}$. Prove that f is measurable if and only if $\{f > \alpha\}$ is measurable for each $\alpha \in S$.

Proof

First suppose f is measurable. Then, by the definition of a measurable function, $\{f > c\}$ is measurable for all $c \in \mathbb{R}$. Since $S \subseteq \mathbb{R}$, it follows that $\{f > \alpha\}$ is measurable for all $\alpha \in S$.

Next, suppose that $\{f > \alpha\}$ is measurable for each $\alpha \in S$. Let $c \in \mathbb{R}$. Since S is dense in $\mathbb{R}$, there exists $s_n \in S$ such that $c < s_n < c + \frac{1}{n}$, for each $n \in \mathbb{N}$. We claim that

$$\{f > c\} = \bigcup_{n=1}^{\infty} \{f > s_n\}.$$

To show this, first let $a \in \bigcup_{n=1}^{\infty} \{f > s_n\}$. Then, for some $N \in \mathbb{N}$, we have $a \in \{f > s_N\}$. This implies that $f(a) > s_N$. But $s_N > c$, so $f(a) > c$. Thus, $a \in \{f > c\}$. Next, let $b \in \{f > c\}$. Then $f(b) > c$. By the Archimedean Property of the real numbers, there exists a natural number N such that $1 < N[f(b) - c]$. Rearrangement of this inequality yields $\frac{1}{N} + c < f(b)$. But $s_N < \frac{1}{N} + c$, so $f(b) > s_N$. Thus, $b \in \{f > s_N\}$, which implies that $b \in \bigcup_{n=1}^{\infty} \{f > s_n\}$. This proves our claim. Now, by hypothesis $\{f > s_n\}$ is measurable for each $n \in \mathbb{N}$ (since each $s_n \in S$). This makes $\{f > c\}$ a countable union of measurable sets. Therefore, it is measurable, as required.

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(4.)

Statement

Let $f : E \rightarrow \mathbb{R}$ be an increasing function, with E measurable. Then f is measurable.

Proof

For each $n \in \mathbb{N}$, define $g_n : E \rightarrow \mathbb{R}$ by $g_n(x) = f(x) + \frac{x}{n}$. We first show that each g_n is strictly increasing. Let $x, y \in \mathbb{E}$ with $x < y$. Then, since f is increasing, we have $f(x) \leq f(y)$. Furthermore, $\frac{x}{n} < \frac{y}{n}$. From these two inequalities, we can conclude that $g_n(x) = f(x) + \frac{x}{n} < f(y) + \frac{y}{n} = g_n(y)$, which means g_n is strictly increasing, as required. Now, since each g_n is strictly increasing, it follows that the sets $\{g_n = c\}$ contain at most one element each. This means they are measurable. Thus, by Proposition 1, each g_n is measurable. Next, we show that the g_n converge pointwise to $f(x)$. Let $\varepsilon > 0$ and let $x \in E$. Choose N to be the smallest integer greater than $\frac{|x|}{\varepsilon}$. Then, whenever $n > N$, we have

$$\begin{aligned} |g_n(x) - f(x)| &= \left| f(x) + \frac{x}{n} - f(x) \right| \\ &= \frac{|x|}{n} \\ &< \frac{|x|}{\frac{|x|}{\varepsilon}} \\ &= \varepsilon. \end{aligned}$$

We have now shown that $\{g_n\}$ is a sequence of measurable functions on E that converges pointwise to f on E . Therefore, by Proposition 9, we conclude that f is measurable. ■